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Fake News Detection: Exploring the Efficiency of Soft and Hard Voting Ensemble

Arifur Rahman^{a,*}, Sakib Zaman^a, Shahriar Parvej^a, Pintu Chandra Shill^a, Md. Shahidul Salim^a, Dola Das^a

^aDepartment of Computer Science and Engineering, Khulna University of Engineering & Technology, Khulna 9203, Bangladesh

Abstract

Fake news dissemination is a critical problem, jeopardizing the reliability of information and eroding public confidence. Machine learning provides a solution by distinguishing between real and misleading news. In this study, we utilized various classifiers such as KNN, Logistic Regression, Decision Tree, LightGBM, XGBoost, Gradient Boosting, Random Forest, and Naive Bayes on an open-source dataset. Tf-idf (Term frequency-inverse document frequency) and Count vectorization were used for text preprocessing, and soft and hard voting ensembles were applied to the best three and five models to boost performance. Additionally, we employed BERT (Bidirectional Encoder Representations from Transformers) and CNN (Convolutional Neural Network) models with various optimizers (adam, rmsprop, SGD, adadelta, and adagrad) to further improve classification. The BERT model, optimized with adam, achieved the highest accuracy of 99.93%, with perfect precision (100%) and high recall (99.84%). Furthermore, using Count vectorization for feature extraction, the soft voting ensemble of the top five models (LGBM, XGB, RF, DT, LR) achieved peak performance among all ensemble models, achieving 99.87% accuracy, with precision, recall, f1-score, MAE, MSE, RMSE, RAE, and RRSE of 99.83%, 99.88%, 99.86%, 0.13%, 0.13%, 3.66%, 0.27%, and 7.80%, respectively. The exhaustive experimentation confirms the applicability and efficiency of the recommended models in identifying fake news.

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Keywords: Fake news detection; Classifier models; Ensemble model; BERT model and CNN model;

1. Introduction

The importance of effective fake news detection in today's complex information ecosystem cannot be overstated. In an era defined by rapid data dissemination and digital interconnectedness, the unchecked propagation of misinformation poses a profound threat to societal stability, democratic processes, and individual decision-making. In the wake of groundbreaking internet advancements and advanced mobile gadgets, alongside the effortless propagation of news across social media arenas, people have shifted towards obtaining information from the web instead of traditional TV stations [1],[2]. Nonetheless, this shift in news consumption bears a contradiction, the authenticity of online news is relatively subpar compared to that of well-established television networks. According to the analysis conducted by the Pew Research Center [3], it was reported that in the month of August in the year 2017, a significant proportion of

* Corresponding author. Tel.: +8801879220747.

E-mail address: rarifkhan652@gmail.com

the American population, totaling 67%, acquired novel information through the utilization of social media platforms. An illustrative case, as reported by a single tweet disseminating rumors, detailed the assertion of “Two explosions in the white house and Barack Obama is injured” during the year 2013. Nevertheless, that baseless speculation achieved its pinnacle of popularity within just six minutes [4]. On August 25, 2015, the baseless rumor related to “shootouts and abductions by drug gangs occurring in the vicinity of Veracruz schools” spread through Twitter and Facebook, triggering widespread turmoil in the city and leading to 26 vehicular collisions as individuals abandoned their cars on the highway to rush and pick up their children from school [16]. In today’s context, social media serves as a potent instrument for journalists and everyday individuals alike, as highlighted by its impact on news propagation [5]. Information regarding breaking news is shared by an originator without undergoing confirmation during the initial posting. Following this, the authenticity of the breaking news can be substantiated to determine its validity or misinformation [6]. The expression denoting news content that remains unconfirmed during its posting is “rumor”. Recent scholarly analyses have portrayed rumors as an “accumulation of details perceived as deceptive” [7]. There are numerous factors accessible for characterizing the nature of a rumor, encompassing elements such as the veracity attribute (accurate, inaccurate, and undetermined) [8]. The comprehensive methodology structure integrates numerous distinct procedures. The whole study was split into seven different subsystems. Section 2 depicts the previous studies on this domain. Section 3 presents the materials and methods employed in this study. The voting ensemble strategies were discussed in section 4. Section 5 describes the proposed CNN model and BERT models architecture. Besides, section 6 presents a detailed result analysis and discussion for the methods applied in this research. Finally, Section 7 concludes the study.

2. Literature Review

Several studies have reviewed cutting-edge advancements in fake news identification, yet few have explored the implications of using both soft and hard voting ensembles on top-performing models as this study does. Alkhodair et al. [9] addressed two key challenges, primarily focusing on retrieving rumor-related microblogs from the internet. Granik et al. [1] proposed a simple approach using the Naïve Bayes algorithm for fake news classification. Bharadwaj et al. [10] used recurrent neural networks, Naïve Bayes, and random forest classifiers, tested on an open-source Kaggle dataset. Chen et al. [11] developed a deep learning model with an attention mechanism to detect rumors in tweets. The importance of user features has also been explored in studies by Liang et al. (2015) [12], Zubiaga et al. (2016) [13], and Castillo et al. (2011) [14]. Mondal et al. (2018) [15] introduced a probabilistic model for early-stage rumor detection during disasters, incorporating key features from the 2015 Chennai flood. Castillo et al. (2011) [14] extracted various features, such as content-based, user-based, and propagation-based, to build a classifier that distinguishes the credibility of microblog posts. Chen et al. (2018) [16] developed a framework using a recurrent neural network (RNN) and autoencoder to understand user behavior. Ma et al. (2016) [17] designed an RNN model to predict the credibility of social media posts by automatically learning features. Liang et al. (2015) [18] classified rumors on Sina Weibo using machine learning and eleven different linguistic and user characteristics. Qazvinian et al. (2011) [19] used three distinct feature categories—content-based, network-based, and microblog-specific memes—to detect both disinformers and those propagating rumors. Liu et al. (2017) [20] approached rumor detection with a diffusion-oriented method, deriving attributes from Sina Weibo messages and using an SVM classifier to classify messages as rumors or non-rumors. This collective research highlights the range of techniques used in rumor and fake news detection but lacks an emphasis on ensemble methods like this study.

3. Materials and methods

3.1. Dataset Description

In this study, we used the well-known ISOT benchmark dataset for fake news detection, accessible via this link: <https://archive.ics.uci.edu/dataset/228/sms+spam+collection>. The dataset consists of two CSV files with categorized articles, as detailed in Table 1. With a balanced distribution, the dataset supports robust model training and testing. The word cloud diagrams of true and fake news can be seen in Figure 1a and Figure 1b respectively. The WordCloud module, typically from the wordcloud library, generated the visual representations of text data where words appear in varying sizes based on their frequency within the text.

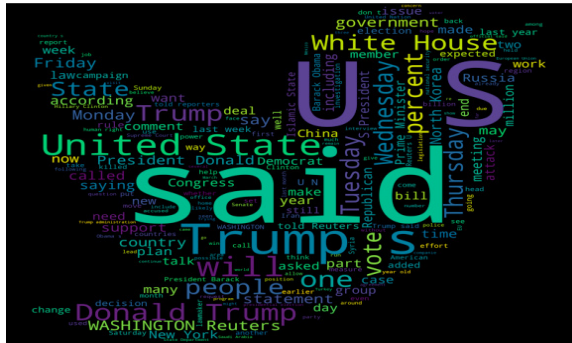
3.2. Dataset preprocessing

3.2.1. Class Level Integration & Datasets Concatenation

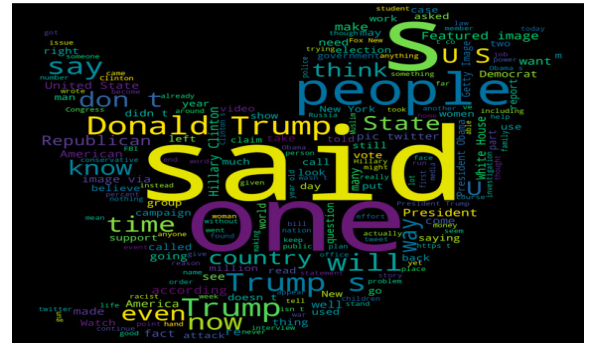
Each record of “True.csv” was endowed with class label of 1. On the other hand, each record of “Fake.csv” was assigned with class label of 0. Then, both “True.csv” & “Fake.csv” dataframes were concatenated. Afterwards, we emphasized on the core attributes of the data by shedding unnecessary attributes.

Table 1. Frequency distribution of different categories for both True.csv and Fake.csv

News	Count	Subjects		News	Count	Subjects	
Fake-news	23,481	Type	Articles size	Real-news	21,417	Type	Articles size
		Middle-east	778			World content	10,145
		Left-news	4,459			Political content	11,272
		Government-news	1,570				
		Political news	6,841				
		US-news	783				
		News	9,050				



(a)



(b)

Fig. 1. (a) Wordcloud visualization of true news corpus. (b) Wordcloud visualization of fake news corpus.

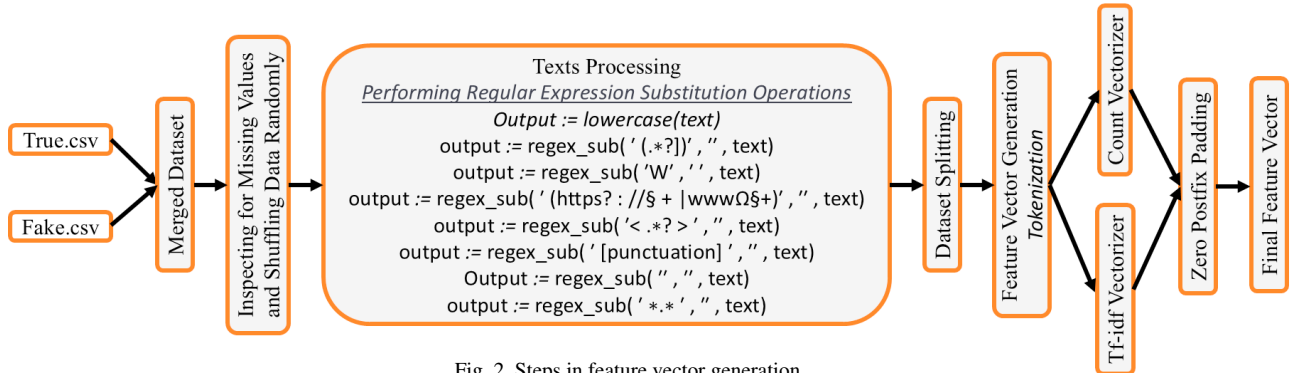


Fig. 2. Steps in feature vector generation.

3.2.2. Processing texts

The merged dataset's input texts were first converted to lowercase, followed by regular expression substitution using the "word_operation" function to ensure consistent processing. Mathematically, the function "word_operation" can be portrayed as follows:

word_operation(text):

Perform Regular Expression Substitution Operation

return *cleaned_text*

Regular expressions, represented as "\[.?\]", "\W", "https?://\S+|www.\S+", "<.*?>" and "\w* \d\w*", acted as virtuoso conductors for text refinement. They excised URLs, square brackets, punctuation marks, HTML tags as well as numeric signs, leaving behind only the textual essence. The regular expression substitution operation searches for patterns enclosed in double quotes("\[.?\]", "\W", "https?://\S+|www.\S+", "<.*?>" and "\w* \d\w*") and removes them from the text. By this way, after a series of text-enhancing operations done by Regular Expression, cleaned texts were returned by the "word_operation" function.

3.2.4. Feature vector generation techniques

[illegible]

BERT model- In the field of Natural Language Processing (NLP), pre-trained language models are powerful tools for tasks like text classification and language generation. Before applying TFBertModel, the text data was tokenized

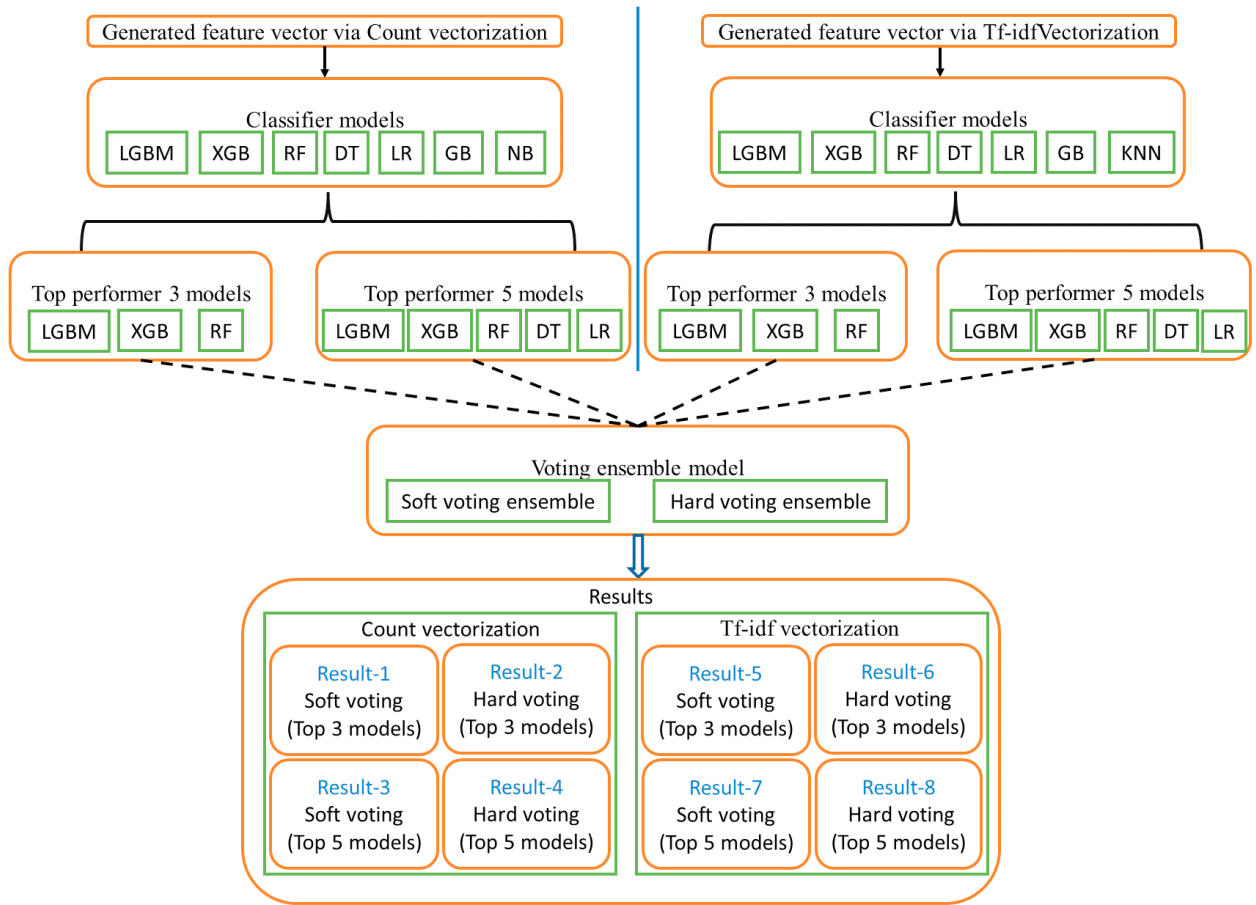


Fig. 3. Workflow of the proposed voting ensemble model.

using BertTokenizer to convert raw text into numeric tokens. These tokens formed feature vectors, which were then inputted into the BERT-based embeddings.

The proposed BERT model begins with an input layer that takes input sequences as integer IDs and input masks. Using the pre-trained BERT model, it extracts contextual embeddings, with the [CLS] token's embedding summarizing the entire input. To prevent overfitting, dropout layers with rates of 20% and 50% are incorporated. Following dropout, dense layers transform the features, with the first dense layer having 64 units and a tanh activation function. The final output layer, designed for binary classification, consists of a single unit with a sigmoid activation function to produce a probability score. The model is compiled using optimizers like Adam, RMSprop, and SGD, with a learning rate of 1×10^{-5} and a binary-cross-entropy loss function.

6. Result analysis and discussion

In this study, model optimization was achieved using various optimizers. For BERT and CNN, Adam consistently improved convergence, achieving high accuracy and stability. Hyperparameter tuning and feature extraction methods, including CountVectorizer and TF-IDF, further enhanced performance by tailoring the models to the dataset.

Classifier models- Tables 2 and 3 present a detailed comparison of various classifier models using Count vectorizer and Tf-idf vectorizer techniques. Our analysis shows that Count vectorization outperformed Tf-idf in most cases, with Light Gradient Boosting Machine (LGBM) achieving 99.84% accuracy compared to 99.73% with Tf-idf. This improvement is attributed to Count vectorization's ability to assign word weights, capture language nuances, and reduce dimensionality, making LGBM the top performer, with precision scores of 99.75% (Count) and 99.65% (Tf-

idf). eXtreme Gradient Boosting (XGB) also performed well, securing 99.76% accuracy with Count vectorization and 99.59% with Tf-idf. Precision, recall, and f1-scores for XGB with Count vectorization were 99.81%, 99.68%, and 99.74%, respectively. Random Forest (RF) ranked third with 99.75% accuracy using Count vectorization, while Decision Tree (DT) ranked third with Tf-idf, scoring 99.58%. Logistic Regression (LR) achieved 99.62% with Count vectorization and 98.60% with Tf-idf. Gradient Boosting (GB) followed, with 99.58% and 99.54% for Count and Tf-idf, respectively. The lowest-performing models were Naive Bayes (NB) and K-Nearest Neighbor (KNN), which scored 93.14% and 78.88% for Count vectorizer and Tf-idf, respectively.

Table 2. Performance comparison of classifier models with Count vectorizer tokenization.

Classifier models	Accuracy (%)	Precision (%)	Recall (%)	F1-val (%)	MAE (%)	MSE (%)	RMSE (%)	RAE (%)	RRSE (%)	AUC (%)
LGBM	99.84%	99.75%	99.90%	99.82%	00.16%	00.16%	4.00%	00.32%	8.54%	99.98%
XGB	99.76%	99.81%	99.68%	99.74%	00.24%	00.24%	4.90%	00.48%	10.46%	99.75%
RF	99.75%	99.64%	99.83%	99.73%	00.25%	00.25%	4.99%	00.50%	10.65%	99.96%
DT	99.71%	99.66%	99.73%	99.69%	00.28%	00.28%	5.34%	00.57%	11.39%	99.71%
LR	99.62%	99.47%	99.73%	99.60%	00.37%	00.37%	6.12%	00.75%	13.05%	99.90%
GB	99.58%	99.22%	99.88%	99.55%	00.42%	00.42%	6.47%	00.84%	13.80%	99.94%
NB	93.14%	88.84%	97.62%	93.02%	6.86%	6.86%	26.20%	13.78%	55.88%	93.81%

Table 3. Performance comparison of classifier models with Tf-idf vectorizer tokenization.

Classifier models	Accuracy (%)	Precision (%)	Recall (%)	F1-val (%)	MAE (%)	MSE (%)	RMSE (%)	RAE (%)	RRSE (%)	AUC (%)
LGBM	99.73%	99.65%	99.79%	99.72%	00.27%	00.27%	5.17%	00.53%	10.79%	99.97%
XGB	99.59%	99.63%	99.52%	99.57%	00.41%	00.41%	6.40%	00.82%	13.36%	99.59%
DT	99.58%	99.67%	99.47%	99.57%	00.42%	00.42%	6.47%	00.84%	13.30%	99.57%
GB	99.54%	99.33%	99.70%	99.52%	00.46%	00.46%	6.81%	00.93%	14.21%	99.87%
RF	99.27%	99.24%	99.24%	99.24%	00.73%	00.73%	8.55%	1.46%	17.84%	99.96%
LR	98.60%	98.31%	98.77%	98.54%	1.40%	1.40%	11.83%	2.80%	24.69%	99.89%
KNN	78.88%	92.38%	60.94%	73.44%	21.12%	21.12%	45.96%	42.32%	95.92%	78.16%

Table 4. Performance comparison of ensemble models with Count vectorizer tokenization.

Ensemble models	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	MAE (%)	MSE (%)	RMSE (%)	RAE (%)	RRSE (%)
Soft voting top 5 (LGBM+XGB+RF+DT+LR)	99.87%	99.83%	99.88%	99.86%	00.13%	00.13%	3.66%	00.27%	7.80%
Hard voting top 5 (LGBM+XGB+RF+DT+LR)	99.85%	99.75%	99.92%	99.84%	00.15%	00.15%	3.89%	00.30%	8.30%
Soft voting top 3 (LGBM+XGB+RF)	99.83%	99.79%	99.85%	99.82%	00.17%	00.17%	4.11%	00.34%	8.78%
Hard voting top 3 (LGBM+XGB+RF)	99.84%	99.75%	99.90%	99.83%	00.16%	00.16%	4.00%	00.32%	8.54%

Voting ensemble (Count vectorization)- In Table 4, the performance of ensemble models with soft voting, using the top five base models (LGBM, XGBoost, RF, DT, LR), achieved a remarkable 99.87% accuracy in fake news classification, with high precision (99.83%), recall (99.88%), and F1-score (99.86%). Hard voting resulted in slightly lower accuracy at 99.85%. Reducing the ensemble to the top three models (LGBM, XGBoost, RF) still maintained impressive results, with soft voting achieving 99.83% accuracy, and hard voting achieving 99.84%. This analysis highlights the stability of both voting strategies and efficiency of using top three models without significant performance loss.

Voting ensemble (Tf-idf vectorization)- From Table 5, the soft voting ensemble with the top 3 models (LGBM, XGBoost, DT) achieved commendable results, with 99.69% accuracy, 99.65% precision, 99.70% recall, and 99.67%

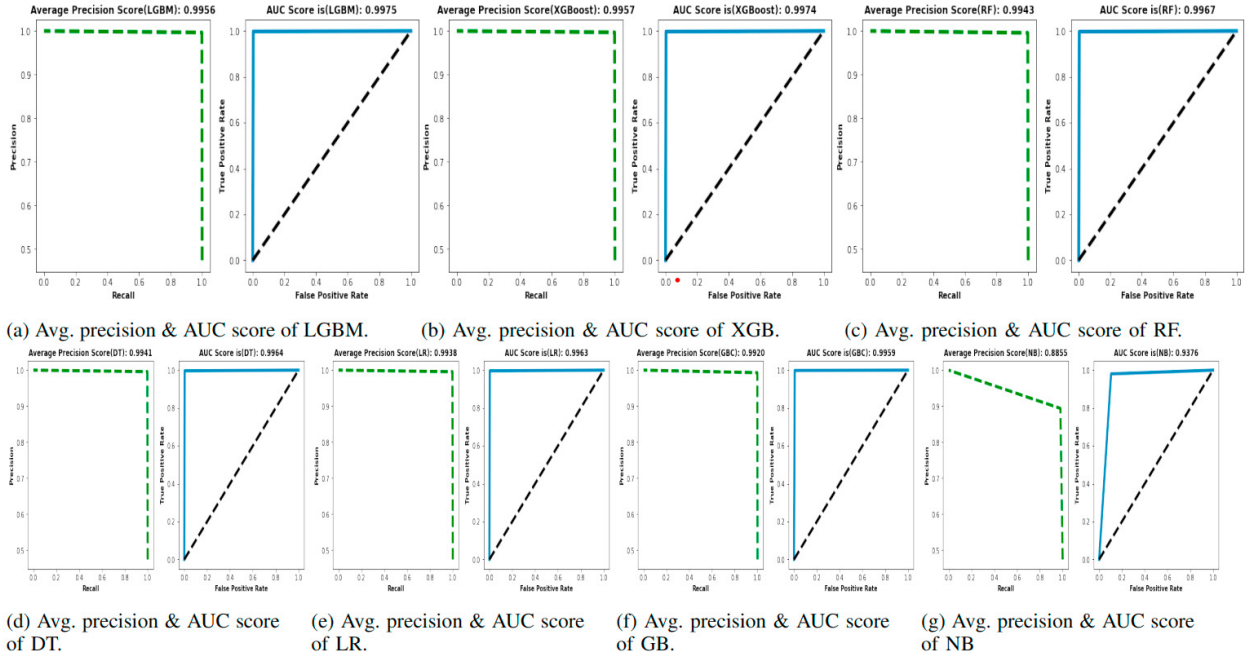


Fig. 4. Graphical visualization of Avg. precision and AUC score for the classifier models with CountVectorizer (CV) tokenization.

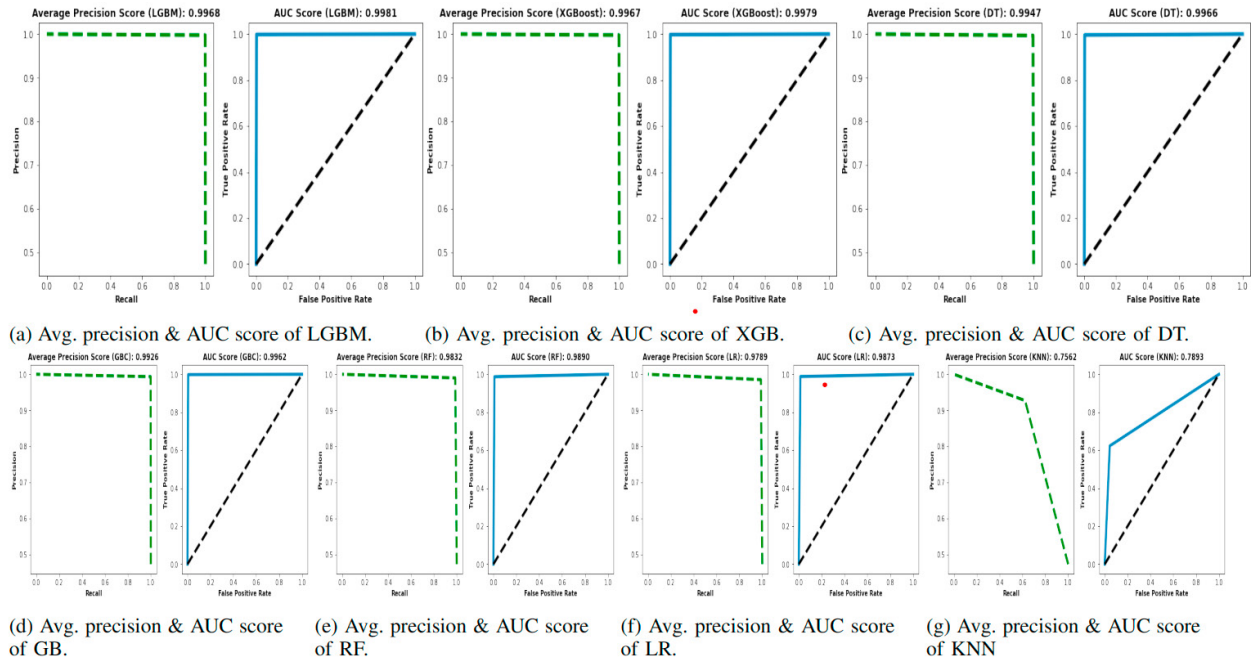


Fig. 5. Graphical visualization of Avg. precision and AUC score for the classifier models with Tf-idfVectorizer (Tf-idf) tokenization.

F1-score. Hard voting performed similarly, with 99.71% accuracy. When using the top 5 models (LGBM, XGBoost, DT, GB, RF), soft voting reached 99.72% accuracy, and hard voting peaked at 99.76%. Both strategies maintained high precision, recall, and F1-scores. The ensembles with Tf-idf tokenization showed consistent and robust performance, confirming the effectiveness of both soft and hard voting, whether using the top 3 or top 5 models, for accurate fake news classification.

Bert model- In Table 6, we compared the BERT model's performance using different optimizers (Adam, RMSprop,

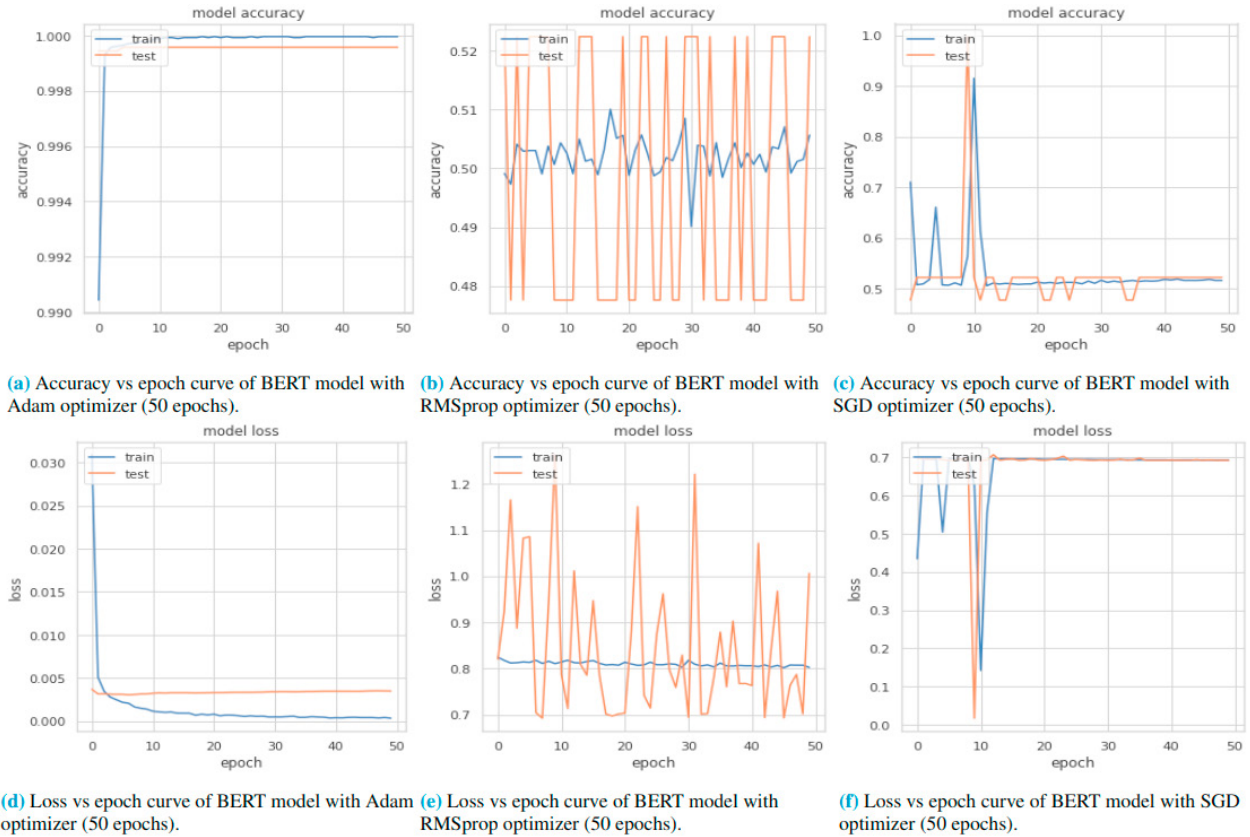


Fig. 6. Accuracy vs epoch and loss vs epoch curves of BERT model with various optimizers (Adam, RMSprop, and SGD) against 50 epochs.

Table 5. Performance comparison of ensemble models with Tf-idf vectorizer tokenization.

Ensemble models	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	MAE (%)	MSE (%)	RMSE (%)	RAE (%)	RRSE (%)
Soft voting top 5 (LGBM+XGB+DT+GB+RF)	99.76%	99.68%	99.74%	99.71%	00.28%	00.28%	5.26%	00.55%	10.97%
Hard voting top 5 (LGBM+XGB+DT+GB+RF)	99.72%	99.70%	99.79%	99.75%	00.24%	00.24%	4.90%	00.48%	10.24%
Soft voting top 3 (LGBM+XGB+DT)	99.69%	99.65%	99.70%	99.67%	00.31%	00.31%	5.58%	00.62%	11.66%
Hard voting top 3 (LGBM+XGB+DT)	99.71%	99.66%	99.74%	99.70%	00.28%	00.28%	5.34%	00.57%	11.14%

and SGD). The BERT model, optimized with Adam, achieved the best results with 99.93% accuracy, 100% precision, 99.84% recall, and a 99.92% F1-score, showing a strong balance between precision and recall. Conversely, RMSprop and SGD led to poor performance, with accuracy dropping to around 51%, and undefined precision and F1-scores. This analysis highlights critical importance of optimizer selection, with Adam emerging as the optimal choice for BERT in fake news detection tasks. Besides, the model was trained for 50 epochs, where accuracy and loss curves plateaued, as shown in Fig. 6.

CNN model- Table 7 compares CNN models with different optimizers (Adadelata, Adagrad, Adam) using Count vectorizer tokenization. The Adam optimizer delivered the best performance, achieving 74.38% accuracy, precision of

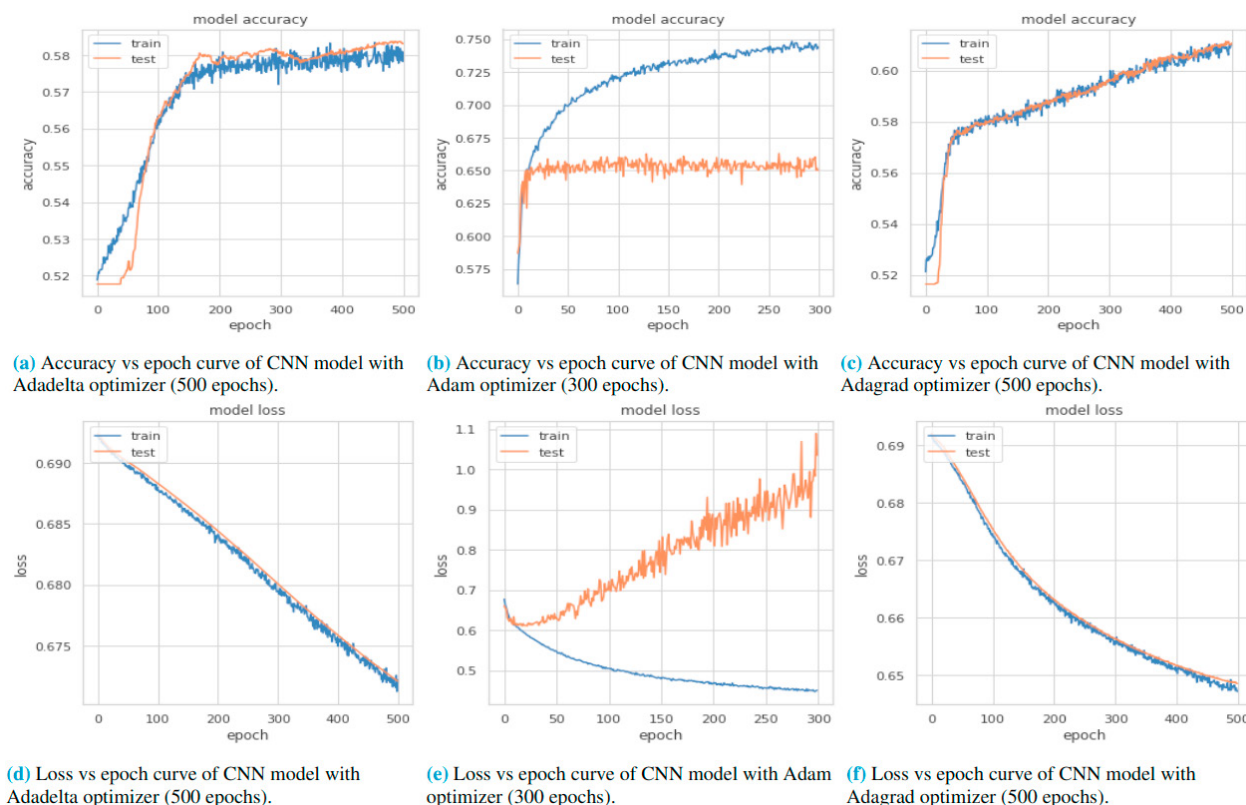


Fig. 7. Accuracy vs epoch and loss vs epoch curves of CNN model with various optimizers (Adadelata, Adam, and Adagrad).

Table 6. Performance comparison of BERT model with various optimizers.

Vectorizer	BERT + Optimizers	Acc.(%) (50 epochs)	Precision (%)	Recall (%)	F1-score (%)
BertTokenizer	BERT + Adam	99.93%	100%	99.84%	99.92%
	BERT + RMSprop	51.02%	undefined	00.00%	undefined
	BERT + SGD	51.97%	undefined	00.00%	undefined

Table 7. Performance comparison of CNN model with various optimizers.

Vectorizer	CNN + Optimizers	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Count vectorizer	CNN+Adadelata	58.08 (500 epochs)	58.69%	37.80%	46.11%
	CNN + Adam	74.38(300 epochs)	66.56%	55.56%	60.57%
	CNN + Adagrad	60.86 (500 epochs)	61.98%	47.83%	54.12%

66.56%, and an F1-score of 60.57%. Adadelata and Adagrad yielded lower results with accuracies of 58.08% and 60.86%, respectively. The optimizer choice significantly impacts CNN performance, with Adam emerging as the most effective. The CNN model was trained for 500 epochs with Adadelata and Adagrad, as accuracy improvements plateaued by then. Both optimizers showed flattened accuracy and loss curves around 500 epochs, while the Adam optimizer reached stable accuracy after 300 epochs, as shown in Fig. 7.

7. Conclusion

This study highlights the influence of models, ensemble techniques, vectorization methods, and optimizers on fake news detection performance. The choice of model and optimizer plays a critical role in accuracy, with the Adam optimizer proving highly effective for both BERT and CNN models, as it helps the models converge efficiently in deep learning tasks. The performance of soft and hard voting ensembles remained consistent, with negligible differences in accuracy, indicating reliability and computational efficiency even when reducing from the top 5 to the top 3 models. Additionally, Count vectorizer showed a slight advantage over Tf-idf due to better feature capturing capabilities. Overall, the results demonstrate that model choice, voting strategies, and vectorization methods significantly impact performance, while Adam stands out as the optimal optimizer. The results underscore the practical implications for real-world application in news verification and social media monitoring. Implementing this model could streamline efforts to identify fake news in large datasets, aiding media organizations and social platforms

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